Automatic Regulation Mechanism in the "Limits to Growth" Archetype

Dreamland Incorporation
Independent Research Division
“Understanding systems before we change them”

Abstract

This paper examines the automatic regulation mechanism within the "Limits to Growth" archetype, using the taxi market as an example. Four scenarios are considered: constant demand, step-by-step demand changes, the introduction of a driver entry delay time, and periodic demand changes.

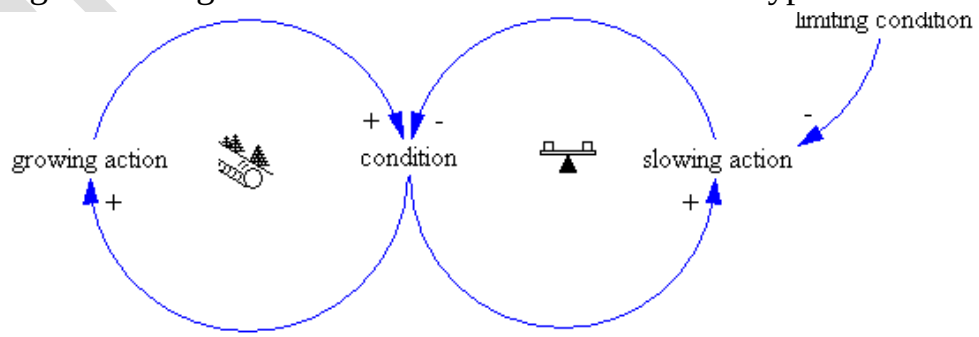
It is shown that the equilibrium number of taxis is determined not by technical capacity, but by the number of passengers. The system automatically reaches equilibrium through a Job attractiveness mechanism defined as the ratio of actual income to target income.

Changes in demand shift the equilibrium, while a delay in decision-making reduces the number of taxis and increases income. Periodic demand changes do not destabilize the system.

1. Introduction

The "Limits to Growth" archetype describes a system in which growth eventually encounters a limiting factor. A diagram of the archetype is shown in Figure 1.

Figure 1. Diagram of the "Limits to Growth" archetype



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In its classical interpretation, this archetype is often accompanied by misguided actions that worsen the situation. This paper considers a simplified version — an automatic regulation mechanism that brings the system to equilibrium without managerial intervention.

The purpose of the study is to investigate how a system with limited demand automatically reaches equilibrium. The taxi market is used as an example: the number of taxis is regulated by driver income, while the limiting factor is the number of solvent passengers.

2. Model Description

Let's assume that at some point in time there are about 1000 solvent passengers willing to use taxi services. If too few taxis are on the road, the fare rises, attracting more drivers. If too many taxis are on the road, the fare falls, and some drivers leave. Thus, the system tends toward an equilibrium determined not by the number of taxis but by the number of passengers. Without passengers, taxis are not needed.

To investigate the behavior of key parameters — number of cars, number of passengers, and hourly income per car — a series of experiments is conducted under the following scenarios:

1) No change in parameters (constant demand);

2) Step-by-step changes in the number of passengers;

3) Step-by-step changes in the number of passengers combined with a driver entry delay time;

4) Periodic changes in the number of passengers.

Excel was used for calculations. Insight Maker was used for model visualization and result verification. The working model diagram is shown in Figure 2.

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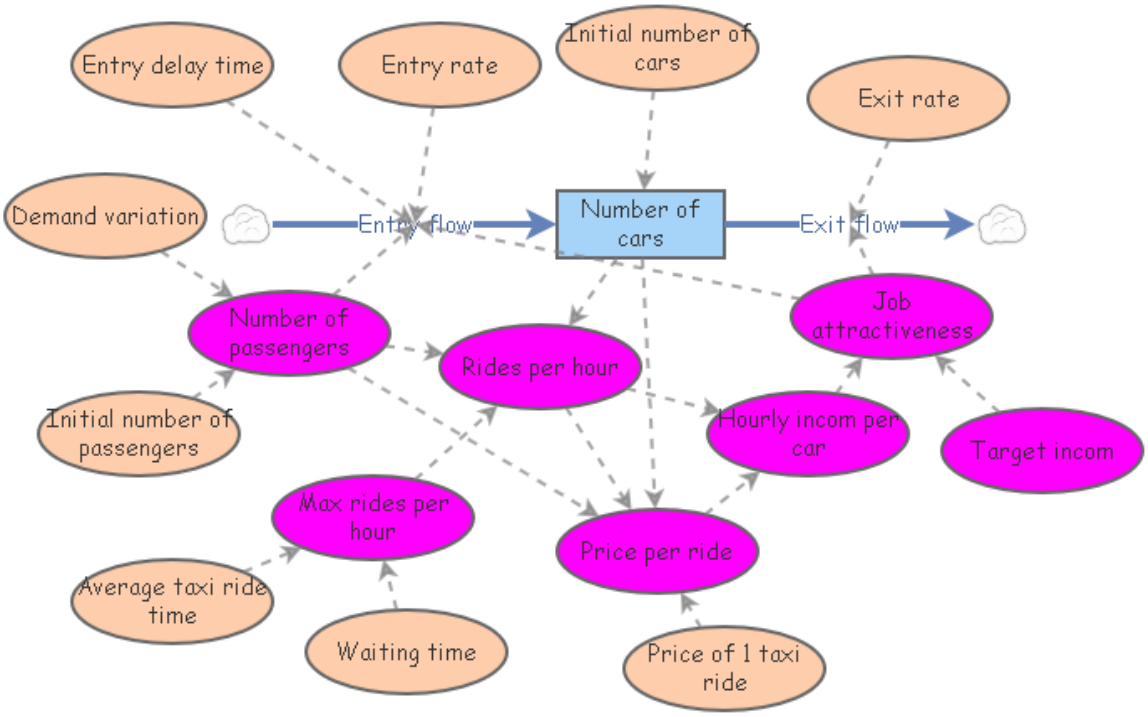
Figure 2. Working model diagram in Insight Maker



2.1. Experiment with Constant Demand

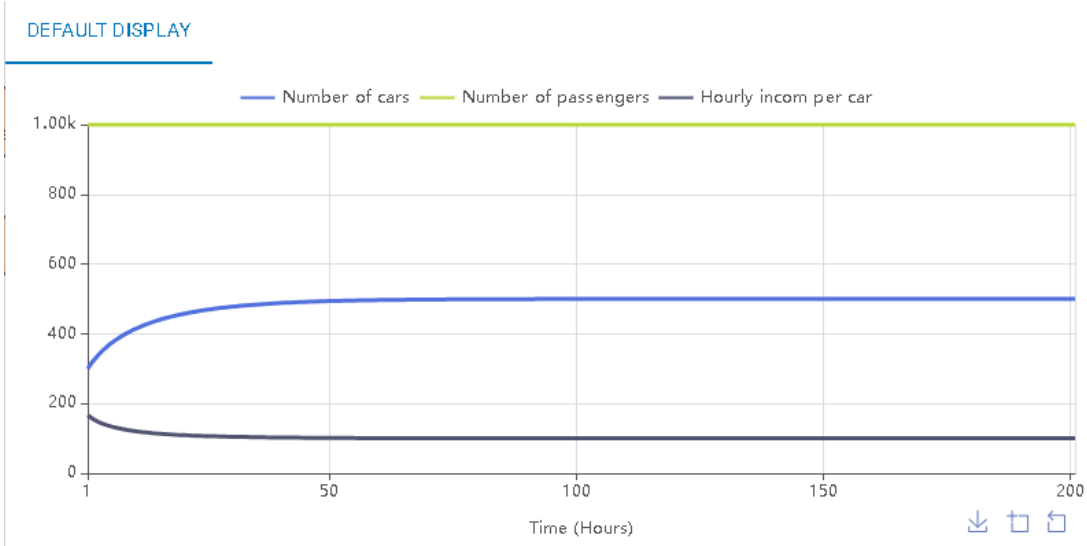
The results of the model with constant demand for the first 15 steps are shown in Figure 3; the graph over the entire time interval (200 steps) is shown in Figure 4.

Figure 3. Model results with constant demand

DEFAULT DISPLAY

Time	Entry flow	Number of cars	Exit flow	Number of passenger	Rides per hour	Price per ride	Hourly incom per car	Job attractiveness
1	20	300	0	1,000	3	55.5555556	166.666667	1.66666667
2	16.875	320	0	1,000	3	52.0833333	156.25	1.5625
3	14.5269017	336.875	0	1,000	2.96846011	50	148.423006	1.48423006
4	12.6861663	351.401902	0	1,000	2.84574442	50	142.287221	1.42287221
5	11.1988234	364.088068	0	1,000	2.74658822	50	137.329411	1.37329411
6	9.96942166	375.286891	0	1,000	2.66462811	50	133.231406	1.33231406
7	8.93511799	385.256313	0	1,000	2.59567453	50	129.783727	1.29783727
8	8.05257756	394.191431	0	1,000	2.5368385	50	126.841925	1.26841925
9	7.29079783	402.244009	0	1,000	2.48605319	50	124.302659	1.24302659
10	6.62692345	409.534806	0	1,000	2.4417949	50	122.089745	1.22089745
11	6.04367948	416.16173	0	1,000	2.40291197	50	120.145598	1.20145598
12	5.52773051	422.205409	0	1,000	2.36851537	50	118.425768	1.18425768
13	5.06859442	427.73314	0	1,000	2.33790629	50	116.895315	1.16895315
14	4.65790179	432.801734	0	1,000	2.31052679	50	115.526339	1.15526339
15	4.2888778	437.459636	0	1,000	2.28592519	50	114.296259	1.14296259

Figure 4. Dynamics of the number of cars, number of passengers, and hourly income per car with constant demand



Since the number of passengers does not change, the limit is reached immediately, without fluctuations. The influence of the number of cars on hourly income is also evident: the more taxis, the lower the fare.

2.2. Experiment with Step-by-step Changes in the Number of Passengers

The results of the model with stepwise demand changes for the first 15 steps are shown in Figure 5; the graph over 200 steps is shown in Figure 6.

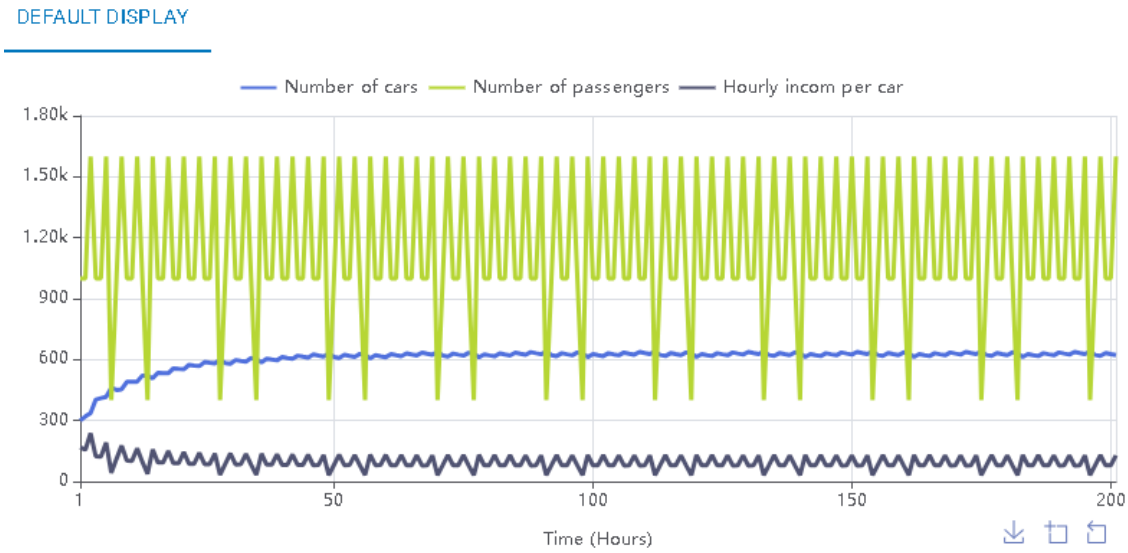
Step-by-step demand change means that demand increases every third step and decreases every seventh step. Otherwise remains at the average level of 1000 passengers.

Figure 5. Model results with step-by-step demand changes

DEFAULT DISPLAY

Time	Entry flow	Number of cars	Exit flow	Number of passenger	Rides per hour	Price per ride	Hourly incom per car	Job attractiveness
1	20	300	0	1,000	3	55.5555556	166.666667	1.66666667
2	16.875	320	0	1,000	3	52.0833333	156.25	1.5625
3	65.9888683	336.875	0	1,600	3	79.1589363	237.476809	2.37476809
4	7.23342097	402.863868	0	1,000	2.48222806	50	124.111403	1.24111403
5	6.57668654	410.097289	0	1,000	2.43844577	50	121.922288	1.21922288
6	44.1583834	416.673976	0	1,600	3	63.9988773	191.996632	1.91996632
7	0	460.832359	10.4332944	400	0.867994602	50	43.3997301	0.433997301
8	3.30379917	450.399065	0	1,000	2.22025328	50	111.012664	1.11012664
9	36.6368913	453.702864	0	1,600	3	58.775619	176.326857	1.76326857
10	0.591033743	490.339755	0	1,000	2.03940225	50	101.970112	1.01970112
11	0.554205063	490.930789	0	1,000	2.036947	50	101.84735	1.0184735
12	30.1305644	491.484994	0	1,600	3	54.2573364	162.772009	1.62772009
13	0	521.615559	0.864622342	1,000	1.91712073	50	95.8560365	0.958560365
14	0	520.750936	12.8300374	400	0.768121519	50	38.4060759	0.384060759
15	27.6023233	507.920899	0	1,600	3	52.5016134	157.50484	1.5750484

Figure 6. Dynamics of the number of cars, number of passengers, and hourly income per car with step-by-step demand changes



Step-by-step demand changes induce oscillations: the system reaches a plateau but does not remain at a constant level as in Experiment 1. When the plateau is reached, a strong dependence of hourly income on the number of passengers is observed: a sharp drop in passenger numbers with a large number of cars leads to a fare reduction.

2.3. Experiment with Driver Entry Delay Time and Step-by-step Demand Changes

The results of the model with a driver entry delay (5 hours) and step-by-step demand changes for the first 15 steps are shown in Figure 7; the graph over 200 steps is shown in Figure 8.

Entering of driver entry delay time means that drivers base their decision on the income that prevailed several hours earlier (in this example, 5 hours ago). They start their shift after demand has already fallen, but they are unaware of it

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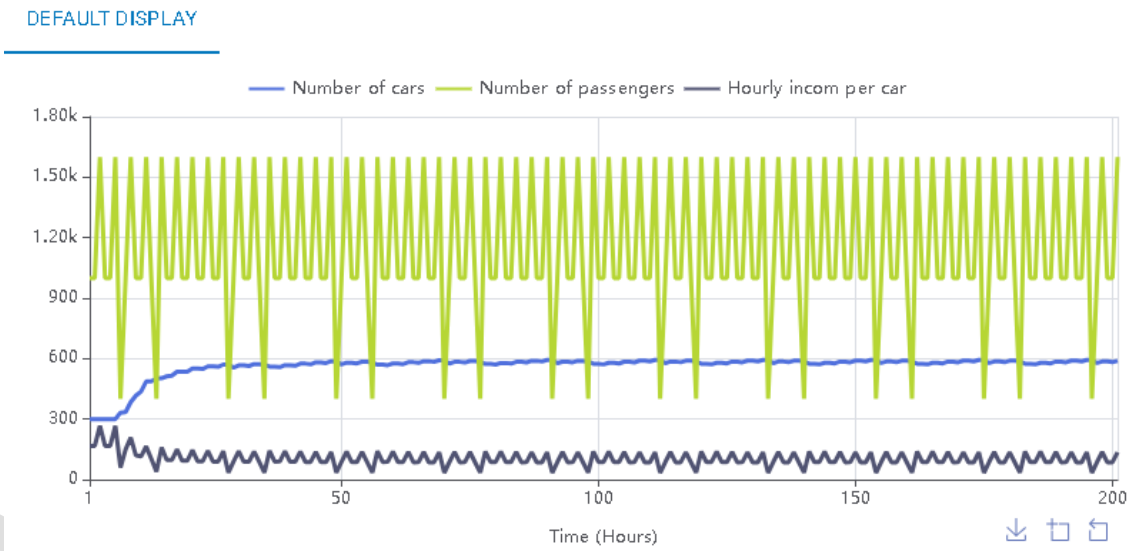
Figure 7. Model results with entry delay time and step-by-step demand changes

DEFAULT DISPLAY

Time	Entry flow	Number of cars	Exit flow	Number of passenger	Rides per hour	Price per ride	Hourly incom per car	Job attractiveness
1	0	300	0	1,000	3	55.5555556	166.666667	1.66666667
2	0	300	0	1,000	3	55.5555556	166.666667	1.66666667
3	0	300	0	1,600	3	88.8888889	266.666667	2.66666667
4	0	300	0	1,000	3	55.5555556	166.666667	1.66666667
5	0	300	0	1,000	3	55.5555556	166.666667	1.66666667
6	32	300	0	1,600	3	88.8888889	266.666667	2.66666667
7	8	332	5.28	400	1.20481928	50	60.2409639	0.602409639
8	50	334.72	0	1,000	2.9875717	50	149.378585	1.49378585
9	32	384.72	0	1,600	3	69.3144798	207.943439	2.07943439
10	20	416.72	0	1,000	2.39969284	50	119.984642	1.19984642
11	50	436.72	0	1,000	2.28979667	50	114.489833	1.14489833
12	0	486.72	0	1,600	3	54.7885163	164.365549	1.64365549
13	14.8135755	486.72	0	1,000	2.05456936	50	102.728468	1.02728468
14	12.9532127	501.533576	12.061343	400	0.797553782	50	39.8776891	0.398776891
15	9.59262814	502.425445	0	1,600	3	53.0758681	159.227604	1.59227604

Figure 8. Dynamics of the number of cars, number of passengers, and hourly income per car with entry delay time and step-by-step demand changes



Despite the same demand as in Experiment 2, the entry delay time leads to fewer drivers and, consequently, to higher income.

2.4. Эксперимент с периодическим изменением количества пассажиров

The results of the model with periodic demand changes for the first 15 steps are shown in Figure 9; the graph over 200 steps is shown in Figure 10.

Periodic demand changes occur every 10 hours as follows:

- Hours 1–10: 1000 passengers
- Hours 11–20: 400 passengers
- Hours 21–30: 1600 passengers
- Hours 31–40: 400 passengers
- Hours 41–50: 1600 passengers, etc.

Figure 9. Model results with periodic demand changes

DEFAULT DISPLAY

Time	Entry flow	Number of cars	Exit flow	Number of passenger	Rides per hour	Price per ride	Hourly incom per car	Job attractiveness
1	20	300	0	1,000	3	55.5555556	166.666667	1.66666667
2	16.875	320	0	1,000	3	52.0833333	156.25	1.5625
3	14.5269017	336.875	0	1,000	2.96846011	50	148.423006	1.48423006
4	12.6861663	351.401902	0	1,000	2.84574442	50	142.287221	1.42287221
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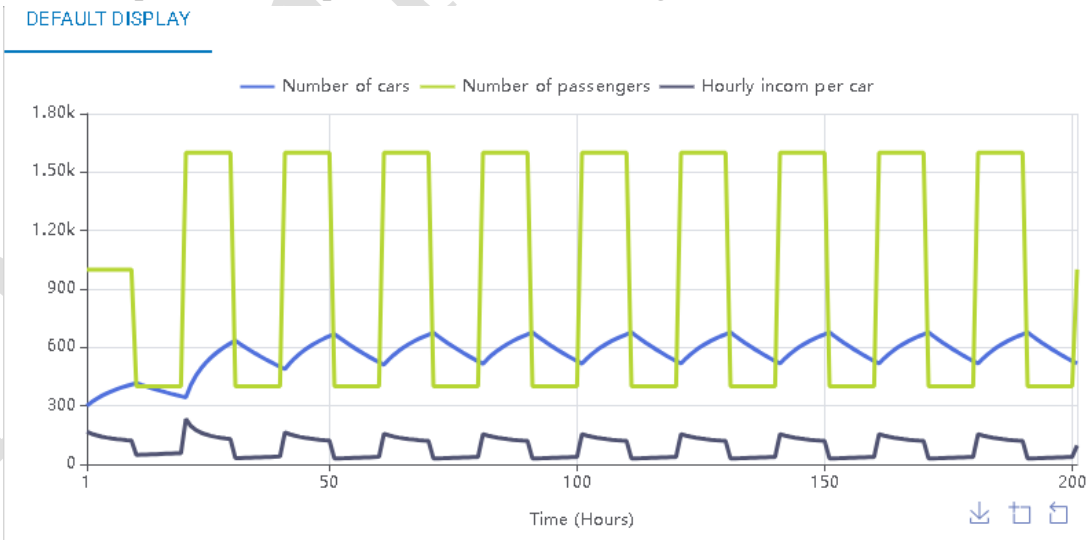
8	8.05257756	394.191431	0	1,000	2.5368385	50	126.841925	1.26841925
9	7.29079783	402.244009	0	1,000	2.48605319	50	124.302659	1.24302659
10	6.62692345	409.534806	0	1,000	2.4417949	50	122.089745	1.22089745
11	0	416.16173	8.64646919	400	0.961164786	50	48.0582393	0.480582393
12	0	407.515261	8.30061042	400	0.981558333	50	49.0779167	0.490779167
13	0	399.21465	7.96858601	400	1.00196724	50	50.0983619	0.500983619
14	0	391.246064	7.64984257	400	1.0223745	50	51.1187251	0.511187251
15	0	383.596222	7.34384886	400	1.04276314	50	52.1381569	0.521381569

Figure 10. Dynamics of the number of cars, number of passengers, and hourly income per car with periodic demand changes



With periodic demand changes, the system also reaches a plateau but does not stabilize at a constant level. The graph clearly shows the dependence of hourly income on the number of passengers: passenger numbers rise first, then the fare increases, and only after that does the number of taxis on the road increase.

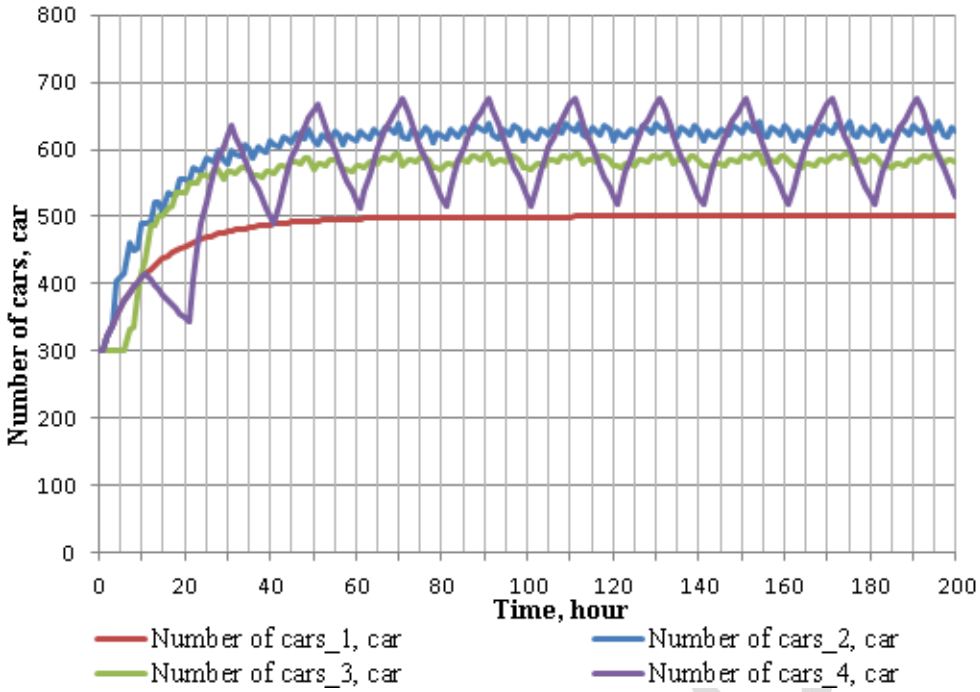
3. Results

Below are comparative graphs for all four experiments. For convenience, all calculations were transferred to Excel. A summary table for the first 30 steps is shown in Figure 11. Figures 12–14 present the dynamics of the number of cars, the number of passengers, and the hourly income per car over the entire time horizon.

Figure 11. Summary table of results for the four experiments

	Entry flow, car/ hour				Number of cars, car				Exit flow, car/ hour				Number of passengers, people				Rides per hour, ride/ hour				Price per ride, $				Hourly incom per car, $/ hour				Job attractiveness			
	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4	Mod 1/	Mod 2/	Mod 3/	Mod 4
0	0,00	0,00	0,00	0,00	300,00	300,00	300,00	300,00	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
1	20,00	20,00	0,00	20,00	300,00	300,00	300,00	300,00	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	3,00	3,00	3,00	3,00	55,56	55,56	55,56	55,56	166,67	166,67	166,67	166,67	1,67	1,67	1,67	1,67
2	16,88	16,88	0,00	16,88	320,00	320,00	300,00	320,00	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	3,00	3,00	3,00	3,00	52,08	52,08	55,56	52,08	156,25	156,25	166,67	156,25	1,56	1,56	1,67	1,56
3	14,53	65,99	0,00	14,53	336,88	336,88	300,00	336,88	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1000,00	2,97	3,00	3,00	2,97	50,00	79,16	88,89	50,00	148,42	237,48	266,67	148,42	1,48	2,37	2,67	1,48
4	12,69	7,23	0,00	12,69	351,40	402,86	300,00	351,40	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	2,85	2,48	3,00	2,85	50,00	50,00	55,56	50,00	142,29	124,11	166,67	142,29	1,42	1,24	1,67	1,42
5	11,20	6,58	0,00	11,20	364,09	410,10	300,00	364,09	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	2,75	2,44	3,00	2,75	50,00	50,00	55,56	50,00	137,33	121,92	166,67	137,33	1,37	1,22	1,67	1,37
6	9,97	44,16	32,00	9,97	375,29	416,67	300,00	375,29	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1000,00	2,66	3,00	3,00	2,66	50,00	64,00	88,89	50,00	133,23	192,00	266,67	133,23	1,33	1,92	2,67	1,33
7	8,94	0,00	8,00	8,94	385,26	460,83	332,00	385,26	0,00	10,43	5,28	0,00	1000,00	400,00	400,00	1000,00	2,60	0,87	1,20	2,60	50,00	50,00	50,00	50,00	129,78	43,40	60,24	129,78	1,30	0,43	0,60	1,30
8	8,05	3,30	50,00	8,05	394,19	450,40	334,72	394,19	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	2,54	2,22	2,99	2,54	50,00	50,00	50,00	50,00	126,84	111,01	149,38	126,84	1,27	1,11	1,49	1,27
9	7,29	36,64	32,00	7,29	402,24	453,70	384,72	402,24	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1000,00	2,49	3,00	3,00	2,49	50,00	58,78	69,31	50,00	124,30	176,33	207,94	124,30	1,24	1,76	2,08	1,24
10	6,63	0,59	20,00	6,63	409,53	490,34	416,72	409,53	0,00	0,00	0,00	0,00	1000,00	1000,00	1000,00	1000,00	2,44	2,04	2,40	2,44	50,00	50,00	50,00	50,00	122,09	101,97	119,98	122,09	1,22	1,02	1,20	1,22
11	6,04	0,55	50,00	0,00	416,16	490,93	436,72	416,16	0,00	0,00	0,00	8,65	1000,00	1000,00	1000,00	400,00	2,40	2,04	2,29	0,96	50,00	50,00	50,00	50,00	120,15	101,85	114,49	48,06	1,20	1,02	1,14	0,48
12	5,53	30,13	0,00	0,00	422,21	491,48	486,72	407,52	0,00	0,00	0,00	8,30	1000,00	1600,00	1600,00	400,00	2,37	3,00	3,00	0,98	50,00	54,26	54,79	50,00	118,43	162,77	164,37	49,08	1,18	1,63	1,64	0,49
13	5,07	0,00	14,81	0,00	427,73	521,62	486,72	399,21	0,00	0,86	0,00	7,97	1000,00	1000,00	1000,00	400,00	2,34	1,92	2,05	1,00	50,00	50,00	50,00	50,00	116,90	95,86	102,73	50,10	1,17	0,96	1,03	0,50
14	4,66	0,00	12,95	0,00	432,80	520,75	501,53	391,25	0,00	12,83	12,06	7,65	1000,00	400,00	400,00	400,00	2,31	0,77	0,80	1,02	50,00	50,00	50,00	50,00	115,53	38,41	39,88	51,12	1,16	0,38	0,40	0,51
15	4,29	27,60	9,59	0,00	437,46	507,92	502,43	383,60	0,00	0,00	0,00	7,34	1000,00	1600,00	1600,00	400,00	2,29	3,00	3,00	1,04	50,00	52,50	53,08	50,00	114,30	157,50	159,23	52,14	1,14	1,58	1,59	0,52
16	3,96	0,00	4,35	0,00	441,75	535,52	512,02	376,25	0,00	1,42	0,48	7,05	1000,00	1000,00	1000,00	400,00	2,26	1,87	1,95	1,06	50,00	50,00	50,00	50,00	113,19	93,37	97,65	53,16	1,13	0,93	0,98	0,53
17	3,65	0,00	19,31	0,00	445,70	534,10	515,88	369,20	0,00	1,36	0,64	6,77	1000,00	1000,00	1000,00	400,00	2,24	1,87	1,94	1,08	50,00	50,00	50,00	50,00	112,18	93,62	96,92	54,17	1,12	0,94	0,97	0,54
18	3,38	24,08	1,31	0,00	449,36	532,74	534,56	362,43	0,00	0,00	0,00	6,50	1000,00	1600,00	1600,00	400,00	2,23	3,00	2,99	1,10	50,00	50,06	50,00	50,00	111,27	150,17	149,66	55,18	1,11	1,50	1,50	0,55
19	3,13	0,00	0,00	0,00	452,74	556,82	535,87	355,94	0,00	2,27	1,43	6,24	1000,00	1000,00	1000,00	400,00	2,21	1,80	1,87	1,12	50,00	50,00	50,00	50,00	110,44	89,80	93,31	56,19	1,10	0,90	0,93	0,56
20	2,90	0,00	17,77	0,00	455,87	554,55	534,43	349,70	0,00	2,18	1,38	5,99	1000,00	1000,00	1000,00	400,00	2,19	1,80	1,87	1,14	50,00	50,00	50,00	50,00	109,68	90,16	93,56	57,19	1,10	0,90	0,94	0,57
21	2,70	21,52	0,00	63,72	458,78	552,36	550,82	343,71	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1600,00	2,18	2,90	2,90	3,00	50,00	50,00	50,00	77,58	108,99	144,83	145,24	232,75	1,09	1,45	1,45	2,33
22	2,50	0,00	0,00	46,25	461,47	573,88	550,82	407,43	0,00	2,96	2,03	0,00	1000,00	1000,00	1000,00	1600,00	2,17	1,74	1,82	3,00	50,00	50,00	50,00	65,45	108,35	87,13	90,77	196,35	1,08	0,87	0,91	1,96
23	2,33	0,00	14,90	36,64	463,98	570,93	548,79	453,68	0,00	2,84	1,95	0,00	1000,00	1000,00	1000,00	1600,00	2,16	1,75	1,82	3,00	50,00	50,00	50,00	58,78	107,76	87,58	91,11	176,34	1,08	0,88	0,91	1,76
24	2,17	19,59	0,00	30,32	466,31	568,09	561,74	490,32	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1600,00	2,14	2,82	2,85	3,00	50,00	50,00	50,00	54,39	107,23	140,82	142,42	163,16	1,07	1,41	1,42	1,63
25	2,02	0,00	0,00	25,76	468,47	587,69	561,74	520,64	0,00	3,51	2,47	0,00	1000,00	1000,00	1000,00	1600,00	2,13	1,70	1,78	3,00	50,00	50,00	50,00	51,22	106,73	85,08	89,01	153,66	1,07	0,85	0,89	1,54
26	1,88	0,00	13,57	22,28	470,49	584,18	559,27	546,39	0,00	3,37	2,37	0,00	1000,00	1000,00	1000,00	1600,00	2,13	1,71	1,79	2,93	50,00	50,00	50,00	50,00	106,27	85,59	89,40	146,41	1,06	0,86	0,89	1,46
27	1,75	18,11	0,00	19,53	472,37	580,81	570,47	568,67	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1600,00	2,12	2,75	2,80	2,81	50,00	50,00	50,00	50,00	105,85	137,74	140,24	140,68	1,06	1,38	1,40	1,41
28	1,64	0,00	0,00	17,28	474,13	598,93	570,47	588,20	0,00	15,96	14,82	0,00	1000,00	400,00	400,00	1600,00	2,11	0,67	0,70	2,72	50,00	50,00	50,00	50,00	105,46	33,39	35,06	136,01	1,05	0,33	0,35	1,36
29	1,53	0,00	12,72	15,42	475,77	582,97	555,65	605,48	0,00	3,32	2,23	0,00	1000,00	1000,00	1000,00	1600,00	2,10	1,72	1,80	2,64	50,00	50,00	50,00	50,00	105,09	85,77	89,98	132,13	1,05	0,86	0,90	1,32
30	1,43	18,25	0,00	13,85	477,29	579,65	566,15	620,90	0,00	0,00	0,00	0,00	1000,00	1600,00	1600,00	1600,00	2,10	2,76	2,83	2,58	50,00	50,00	50,00	50,00	104,76	138,01	141,31	128,84	1,05	1,38	1,41	1,29

Figure 12. Comparison of the number of cars



The smallest number of taxis is observed in the first experiment (pure limits to growth, without demand changes or delays). Introducing demand changes (Experiments 2 and 3) increases the number of cars: the maximum number of passengers shifts from 1000 to 1600 (Figure 13), which leads to a larger vehicle fleet.

In Experiment 3, a driver entry delay time was added. Drivers respond not to current demand but to past income. When demand starts to fall, drivers do not leave immediately, keeping in mind the high earnings of the previous few hours. This keeps taxis on the road until income drops to the target level ($100). As a result, the number of taxis in Experiment 3 is smaller than in Experiment 2.

In Experiment 4 (periodic demand changes), the system reaches a stable plateau. The absence of time delays allows it to react instantly to changes in the number of passengers.

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Figure 13. Comparison of the number of passengers

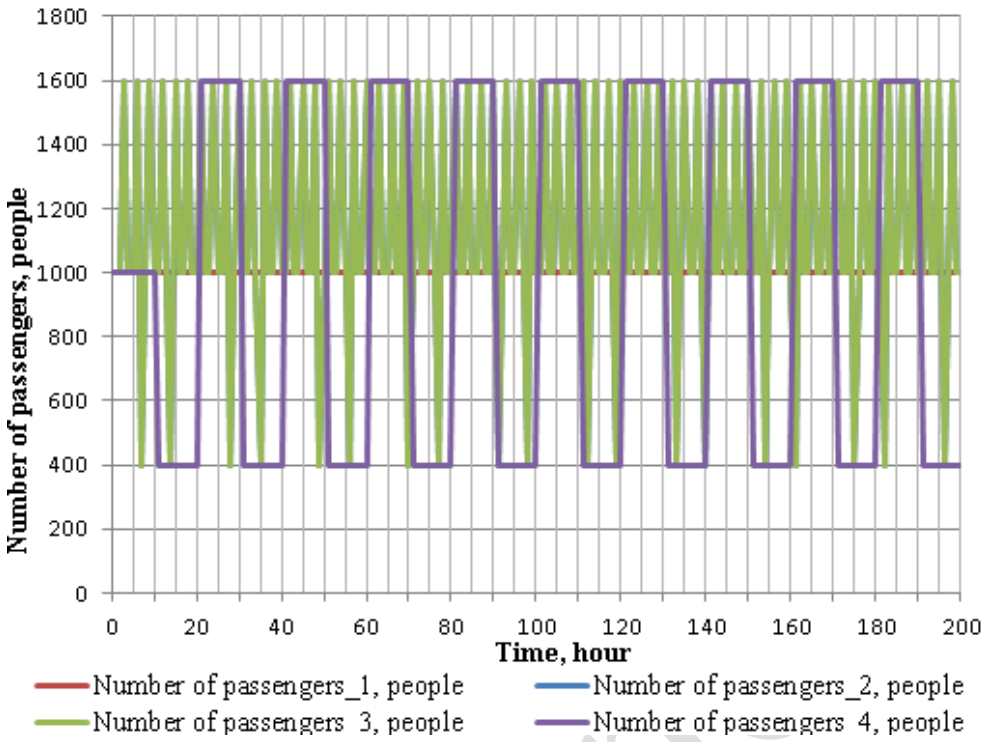
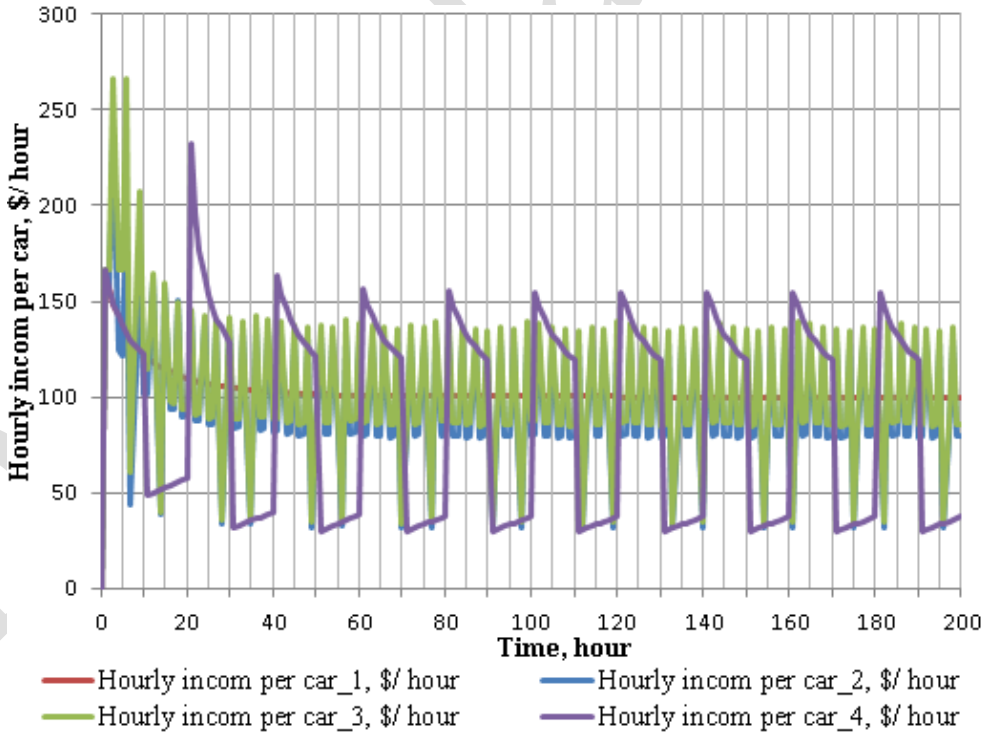


Figure 14. Comparison of hourly income per car



The highest incomes are observed in the first 30 steps, when the system is just reaching equilibrium and has not yet hit the limit determined by the number of passengers. After reaching the plateau, incomes behave predictably and synchronously with the number of cars.

4. Conclusion

This work examined the "Limits to Growth" archetype using the taxi market as an example. Four scenarios were considered: constant demand, step-by-step demand changes, step-by-step demand changes combined with a driver entry delay time, and periodic demand changes.

The main findings are:

1) The limit is determined by the number of passengers. The number of cars is regulated through the job attractiveness mechanism (the ratio of actual income to target income). When there are too few taxis, income rises, attracting more drivers; when there are too many, income falls, and drivers leave. Equilibrium is reached when income matches drivers' expectations (Job attractiveness = 1).

2) Changes in demand shift the equilibrium. When the maximum number of passengers increases from 1000 to 1600, the equilibrium number of taxis also rises. The system moves to a new plateau.

3) A decision-making delay reduces the number of taxis. When drivers base their decisions on past income rather than current demand, they enter the market with a delay. As a result, fewer taxis are on the road, and income is higher than in the scenario without delay.

4) Periodic demand changes do not destabilize the system. The system reacts instantly to fluctuations in passenger numbers and maintains its self-regulating ability.

Thus, in its basic form, the "Limits to Growth" archetype demonstrates an automatic equilibrium mechanism in which the limiting factor is not the number of taxis but an external resource — in this case, the number of solvent passengers.

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